LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Exhaust system, exhaust gas flow, constructional details and materials
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Exhaust system, exhaust gas flow, constructional details and materials
5. Specific Objectives	13.1.2.2 Trainee will be able to explain function and operation of each components of gas turbine engine.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examine engine cutaway model to identify exhaust nozzle, diffuser, and liner; verify safety precautions regarding hot surfaces and rotating parts; instructor supervises identification and material inspection.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the exhaust system?
(b) Name two materials commonly used in exhaust liners.
(c) How does exhaust gas flow direction change after the turbine?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Exhaust gas flow path • Exhaust nozzle function • Thrust generation principle • Exhaust diffuser role • Pressure recovery mechanism	L & Ds	PPT, IB	15 min
(b)	• Exhaust liner construction • Heat resistant alloys • Cooling air circulation • Nozzle guide vanes • Material thermal properties	L & Ds	PPT, PDF	20 min
(c)	• Exhaust system maintenance • Inspection criteria • Material degradation signs • Nozzle adjustment procedures • Safety during inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the exhaust nozzle converts pressure into thrust.
(b) Describe the role of cooling air in exhaust liner construction.
(c) List three signs of material degradation in the exhaust system.

12. Home Assignment and Follow Up	Prepare a brief note on the fuel system pressure control and flow control mechanisms to be discussed in LP 10.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fuel system, pressure control and flow control system, plunger types pump, gear type pump, various stages of fuel atomization and types of burners.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min